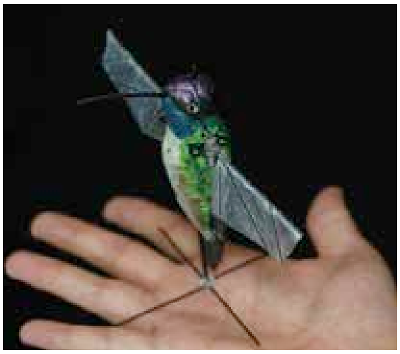
Flappy hummingbird robot goes where drones cannot

One of the envisioned use cases for drones, and even drone swarms is for search and rescue operations. These flying robots can provide on ground surveillance for rescue workers, soon after a disaster situation. Such drones can potentially be used to identify trapped or injured humans for the rescue teams to move in. Quadcopters and fixed wing drones cannot easily maneuver themselves through collapsed buildings and cluttered areas.



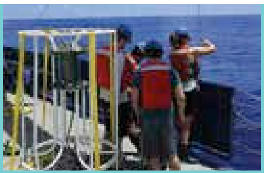
Researchers at Purdue University have built a hummingbird robot, that is trained using AI to autonomously fly around and behave like a real hummingbird would. For example, the robot can learn when the time is right to perform an escape maneuver. The hummingbird bot does not even have imaging sensors, which makes it more lightweight, and reduces the number of subsystems on board. Every time the robot bumps against something, the electrical current inside is disrupted, which is tracked by the robot to understand the area that it is operating in. Xinyan Deng, an associate professor of mechanical engineering at Purdue explains, "The robot can essentially create a map without seeing its surroundings. This could be helpful in a situation when the robot might be searching for victims in a dark place – and it means one less sensor to add when we do give the robot the ability to see."

There is a limit to how small drones can get. The tinier the drones, the heavier and more viscous the air around it gets. The smallest insects practically swim through the air, instead of flying. The air is simply too thick to generate lift with rotor blades or fixed wings. The reason why the researchers used the hummingbird as a model is because of how different the physics of its flapping wings are, "the aerodynamics is inherently unsteady, with high angles of attack and high lift. This makes it possible for smaller, flying animals to exist, and also possible for us to scale down flapping wing robots," Deng explains. The group of researchers that built the hummingbird robot spent years studying hummingbirds in flight, translating the motions into algorithms that could be taught to the robots.

The chassis of the robot is 3D printed. The framework of the wings is made up of high strength but lightweight carbon fibre, which are then covered by laser cut membranes. One robot has the weight of an actual adult hummingbird - 12 grams, which has a lifting capacity of 27 grams. The high lifting capacity of the robots allow them to be loaded with additional sensors. The flight is nearly silent, which is useful for any potential covert operations. Although real hummingbirds have a complex system of muscles powering their wings, the robot version has just one motor per wing. Each wing can be controlled independently of the other. The same team of researchers have also created a tiny flapping wing drone, weighing just 1 gram.
https://dgit.in/birdbot

New Antibiotics

The overuse of antibiotics are making bacteria increasingly resistant, causing a global healthcare crisis. By evolving antibiotics in lab conditions till they get the desired results, researchers can now tackle the increasingly resilient bacteria.
https://dgit.in/evoab



Arsenic breathing life

For most living things, arsenic is poison. Researchers have discovered microorganisms that are metabolising arsenic in the Pacific Ocean. they could be an ancient form of life that existed long before photosynthesis came about.
https://dgit.in/arsbio

Wolverine Bones

Researchers are now 3D scanning bones of patients to 3D print custom titanium bones, that can then be safely implanted within the body. This can help patients who are set to have limbs amputated because of disease or accidents.
https://dgit.in/titbones

Scraps as fuel

Biotechnologists at the University of Waterloo have come up with a way to process table scraps and biological waste, into a substitute for fossil fuel. Most bio waste ends up in landfills, but with the new process, they can be processed in small and medium scales.
https://dgit.in/biofuel